

Full Stack Development with AI: Programme Outline

Modules	Weeks	Date	Module Name	Learning Outcomes
Module 01	week 1	Wednesday, July 8, 2026	Fundamental for beginners	- Grasp programming fundamentals, including languages, algorithms, algorithmic thinking, development environments, variables, data types, operators, control flows, Boolean expressions, and conditional and iterative flows. - Differentiate between static and dynamic web pages. - Understand the learning path to Full Stack Development.
Module 02	week 2	Wednesday, July 15, 2026	Introduction to Git	- Understand version control systems, particularly the advantages of distributed systems like Git. - Learn the basic and remote Git workflows, branching mechanisms, and resolving merge conflicts, encompassing tasks such as creating, switching, and merging branches, handling files, and managing a remote repository.
Module 03	week 3	Wednesday, July 22, 2026	HTML and CSS Fundamentals	- Learn the structure and syntax of HTML, usage of common tags, and types of URLs, along with form-related tags. - Gain knowledge of CSS declaration methods, basic selectors, styling techniques, the DOM, and publishing webpages to the WWW, while integrating HTML and CSS to create webpages.
Module 04	week 4	Wednesday, July 29, 2026	Intermediate HTML and CSS	- Discover the benefits of semantic tags, advanced CSS selectors, the CSS Box Model, and layout techniques using Flexbox and Grid. - Explore responsive design with media queries, frontend design frameworks, CSS frameworks like Bootstrap, and working with CSS templates.

Module 05	week 5	Wednesday, August 5, 2026	Introduction to JavaScript	• Understand JavaScript, its technical characteristics, and high-level programming features, and differentiate between classic and modern JavaScript. • Learn to declare variables, understand data types, dynamic typing, and Boolean expressions, write control flow statements and loops, define and invoke functions, use arrays and objects, explain the DOM programming interface, and manipulate DOM objects.
Module 06	Week 6	Wednesday, August 12, 2026	Introduction to Python—Part 1	• Understand Python, its technical characteristics, and its high-level programming features. • Understand data types, explain dynamic typing, and manipulate values using operators in Python. • Understand control flow statements and write if and switch case statements in Python. • Differentiate between loop types, and use break and continue statements. • Define, invoke, and return values from functions while understanding dynamic typing in Python functions. • Use Python lists, dictionaries, sets, tuples, and multidimensional data structures while comparing their characteristics. • Understand the limitations of Python's basic data structures in storing and manipulating data.
Module 06	Week 7	Wednesday, August 19, 2026	Introduction to Python—Part 2	• Understand the role of common Python libraries in data science and AI workflows, including an overview of NumPy, Pandas, Matplotlib, and scikit-learn. • Understand the structure and technical characteristics of NumPy's ndarray and perform basic data operations. • Perform advanced indexing, slicing, and apply universal functions, conditional logic, statistical calculations, linear algebra, and random number generation with NumPy. • Understand the structure and technical characteristics of Pandas Series and perform basic data operations.

				• Understand the structure and technical characteristics of Pandas DataFrame and perform basic data operations. • Import raw data into a Pandas DataFrame, handle missing data, perform feature engineering, and export processed data. • Understand the overview and architecture of Matplotlib and create basic charts. • Perform exploratory data analysis using Matplotlib, create pivot tables with Pandas, and generate advanced charts.
Module 07	week 8	Wednesday, August 26, 2026	Frontend frameworks	• Recognise the role and importance of frontend frameworks in modern web development, identifying key features and benefits. • Learn about traditional and modern approaches to frontend development, the declarative approach, organizing web applications into components, Single Page Applications (SPAs), and the software requirements for getting started with frontend development.
Module 08	Week 9	Wednesday, September 2, 2026	Working with React— Part 1	• Recognise Learn what React is, its benefits, and how to set up a webpage to work with React, including JSX and transpiling. • Explore how to create and work with React components, including the use of props and evaluating JavaScript expressions. • Gain knowledge of the ES6 Module System and how to organise React applications effectively. • Familiarise yourself with commonly used ES6 constructs in React and how to build a simple React application. • Understand the overview and architecture of Matplotlib and create basic charts. • Perform exploratory data analysis using Matplotlib, create pivot tables with Pandas, and generate advanced charts.
Module 08	Week 10	Wednesday, September 9, 2026	Working with React — Part 2	• Explain what HTML events are, how to handle events in a webpage, and how event handling works in React.

				• Differentiate between props and states, and explore how to use React Hooks to define state variables. • Distinguish between functional and class components in React, and compare Controlled vs. Uncontrolled Components. • Work with JSON, AJAX, and the fetch() API, and understand how to implement React Router and Create React App for building multi-page applications.
Module 09	week 11	Wednesday, September 23, 2026	Relational Databases and SQL	• Explore databases, DBMS, SQL, types, and enterprise relational databases. • Learn data modeling, MySQL installation, relational tables, keys, integrity rules, relationships, Entity-Relationship model, MySQL Workbench, DDL, DML, SQL schemas, commands (INSERT, UPDATE, DELETE, SELECT), WHERE clause, operators, aggregation, SQL joins, and database integration with webserver applications like Node.js in full stack development.
Module 10	week 12	Wednesday, September 30, 2026	Introduction to Backend Development	• Differentiate frontend and backend software engineering, understand server-side vs. client-side rendering. Implement server-side rendering with JavaScript (Node.js) and Python (Flask). • Learn Node.js and Express.js, creating static HTML, routing, handling query strings, HTTP methods, session state, and variables. • Grasp Flask's technical aspects, creating static HTML with Jinja2, and managing routing, query strings, HTTP methods, and session state.
Module 11	week 13	Wednesday, October 7, 2026	Non-relational Databases and MongoDB	• Understand NoSQL Databases versus SQL Databases, explore MongoDB's ecosystem and client scenarios. • Grasp concepts like Documents, Collections, JSON Data, and BSON Syntax. Install MongoDB server and Compass, manage server services, and set up MongoDB Atlas. • Learn to manipulate documents with Mongosh and Mongo shell, master query syntax and operators, handle arrays and nested

				documents, and use Data Aggregation Pipelines. Integrate MongoDB with Node.js using Direct Drivers or ODM libraries, install drivers, and develop applications with MongoDB.
Module 12	week 14	Wednesday, October 14, 2026	RESTful API Development	• Understand Service-Oriented Architecture (SOA) and Microservices Architecture, define software and web services, compare RESTful and SOAP web services focusing on HTTP's role. • Utilise JSON format in JavaScript and Python for data representation, implement best practices and error handling in RESTful API endpoint design, and optimise endpoint performance. • Create CRUD operations with Express.js and Flask/Connexion for RESTful API endpoints, integrate them with databases, and test them using tools like Postman.
Module 13	week 15	Wednesday, October 28, 2026	Introduction to Artificial Intelligence and Machine Learning	• Discover Artificial Intelligence (AI) and its subfields, recent advancements in AI, and the concepts of Supervised Learning (SL), Unsupervised Learning (USL), and Reinforcement Learning (RL) within Machine Learning (ML). • Gain familiarity with Python's Machine Learning libraries.
Module 14	week 16	Wednesday, November 4, 2026	Neural Networks and Deep Learning	• Understand the foundations of Artificial Neural Networks (ANNs), including biological inspirations and motivations. • Understand Deep Learning concepts, including its applications in Computer Vision using Convolutional Neural Networks (CNN) and Natural Language Processing (NLP).
Module 15	week 17	Wednesday, November 11, 2026	AI Libraries and Frameworks	• Gain insight into ANN components such as neurons, activation functions, single-layer and multi-layer architectures, and the overall training process. • Implement a basic Neural Network using the Keras library in Python.

Module 16	week 18	Wednesday , November 18, 2026	Integrating AI Models into Full Stack Applications	• Recognise the distinction between model training, deployment, and concepts of model persistence and serving. • Compare approaches for saving/loading scikit-learn and Keras models, and select the best methods. • Create RESTful API endpoints with Flask and Connexion to serve models, handle input parameters, and file uploads. • Consume these endpoints in React with axios, handle asynchronous processing, and similarly consume endpoints in Express.js and Flask with axios and Requests. • Explore model serving beyond RESTful endpoints using MLflow, create, serve, and consume models with MLflow in Python.
Module 17	week 19	Wednesday, November 25, 2026	Introduction to AI driven features	• Discover the purpose and types of recommendation systems, including content-based and collaborative filtering approaches, and appreciate embedding spaces and similarity measures. • Analyse steps to build content-based and collaborative filtering recommender engines using methods like word embedding tools (Word2vec, fastText, GloVe) and matrix factorisation models, including regularisation and softmax model training.
Module 18	week 20	Wednesday, December 2, 2026	Performance Optimization and Scalability	• Understand the significance of performance optimisation in full stack applications. • Identify scalability challenges and implement appropriate solutions. • Learn techniques for enhancing AI model performance and efficiency. • Implement strategies for scaling applications to handle large volumes of data effectively. • Explore tools and techniques for load testing, performance monitoring, and identifying performance bottlenecks. • Gain insights into real-world examples and best practices for performance optimization and scalability.

Module 19	Week 21	Wednesday, December 9, 2026	Generative AI application— Part 1	• Understand how Generative AI is used in full stack development and its diverse applications. • Using Large Language Models for software development tasks. • Working with prompts, parameters, and fine-tuning for optimising the output of LLM.
Module 19	Week 22	Wednesday, December 16, 2026	Generative AI application— Part 2	• Understand LLM performance, benchmarks, deployment considerations for LLMs and choosing the right model based on use case requirements. • Explore the concept of autonomous AI agents and the pros and cons of using them in software development.
	Weeks 23 and 24	Wednesday, December 23, 2026	Capstone Weeks	Capstone Project